

colour reproduction ability on a photo paper, an ideal output should look as striking and brilliant as that taken from a studio.

All these printers were tested on our test bed, comprising an Intel Pentium 4 2.2 GHz processor seated on an original Intel 845 GEBV2 motherboard, with 256 MB DDR RAM and a 40 GB 7,200 rpm ATA IV hard disk. Windows XP was our operating system. For the graphics sub-system, we used a SiS Xaber 600 graphics card. We used Word XP for printing the word document, whereas Adobe Acrobat Reader 5.0 was used to print the combination document. For printing the image file, we used Adobe Photoshop 7.0. The printers that came with both parallel and USB interface were tested only on the USB port.

How they fared

If you are out in the market trying to purchase an inkjet printer, you will notice that the majority of the vendors will recommend an HP or a Canon as a suitable solution. Such is the cult status that these companies have acquired within the printer market; but there are other printers companies, such as Epson and Lexmark apart from Olivetti and Sharp (which are the least known brands) that are just as good. If you think that a high price indicates the performance of a product, you may be in for some disappointment. The additional price is often more for the extra features, than the speed, hence a lower-priced printer may not have too many features, but can be the fastest on the block. Considering

all these factors, we decided to take them all on a printing spree to find out which of these printers did their job the best.

Home printing solutions

So who is a home user? Exact definitions apart, a home user is one who usually takes one or two odd prints—infrequently. Most of the printing material is textual, with an odd photo print. Hence, price plays a major role in the buying decision, followed by performance and support from the printer manufacturer.

After the relevant scores were gathered, we calculated the Suitability to Home score, which determined the success of a particular printer in this application area. To further narrow down on the ideal solution for this application area, we took into consideration the price. Thus, the printers that cost between Rs 3,000 to Rs 5,500—and scored greater than 65 STTI—are suited to home printing. Based on this, we present the following analysis.

The best printers for home users

For a printer to be suited to a home user, it should have a decent printing speed, with acceptable print quality under both normal and best ink settings. Also, there shouldn't be too much of a time difference between printing text and image documents. The quality of the photo printout should be satisfactory.

The HP 3325 printer came out on top of the home-printing group because of its excellent price to performance ratio. This printer was the least-priced printer in the whole test, and defied the 'higher the price, better the performance' theory.

The HP 3325 took only 18 seconds to print out the text page, and only 169 seconds to print the image document; this feat is amazing when considered that photo printing devices, such as the Canon S530D, took 223 seconds! The final image quality was not sacrificed upon, and all the details were reproduced at a more-than-satisfactory level. Marrying such a fine performance with the low price tag of Rs 2,999 produced the excellent STTI score of 65.79.

The Lexmark twins, Z25 and Z35, that fall in the same price band, showed below par performance. Although both these printers were as good as the HP 3325 at printing black and white text printouts, they failed to catch up when the work demanded colour printing. They took 569 and 570 seconds respectively, to print the image document. All this combined, gave them a low STTI score of 43 and 45, respectively.

The HP 3420 also performed acceptably in all the tests, but lost some valuable points to the speed of printing, by taking longer to print the image file. The only reason it lost on overall grounds at the STTI index was because of its price—a good Rs 1,000 more than the HP 3325. Other printers scored a higher STTI score, in the order of 79, but simply didn't fit the bill. Hence the HP 3325 was the clear winner.

Our recommendation for an ideal home printing solution would be the HP 3325 since it hits the sweet spot with good performance and a price that fits everybody's pocket.

Office printing solution

When office resources such as printers, are shared over a network, the printing speed is of most importance. Features, such as a printer's buffer memory, directly affect its performance—since data transfer rates and queuing time between print jobs come into the picture. Practical features, such as checking the ink-status through the provided driver, can further help a system administrator, as a time-saving device. With USB 2.0 connectiv-

To Market, to Market



- Choose the printer based on the kind of print quality you require—i.e. in general, Inkjet printers offer excellent quality as far as images and graphics are concerned, whereas a laser printer offers excellent text quality printouts. (We haven't tested laser printers in this issue.)
- While buying an inkjet printer, do check the page per minute speed (ppm) since the manufacturer's rated speed is always with context to the printer's fastest mode of printing and not its best quality mode, which is used quite often.
- If you intend to share the printer over the network, look out for the amount of buffer memory—the more the better.
- If you are buying the printer for photo printouts, always take a test print to ensure the required quality. Also remember that most printers require a separate ink cartridge for photo-quality printouts; check this, else you will have to shell out an additional amount for a photo cartridge.
- Check the type of interface supported by your PC, generally all PCs today have USB ports, so buy a printer with a USB interface, as it offers a higher speed of data transfer.
- Consider the total cost of ownership, and not just the purchase price. Depending on how many pages you print, even a small difference of a rupee will save you a sizeable amount of cash over the lifetime of the printer.
- Check whether the printer supports additional features, such as ink status determination via software so that you can change the cartridge upon indication.
- Check the manufacturer's Web site for the type of support they offer in the form of regular software updates, patches, etc.
- Ensure that the footprint of the printer is such that it sits comfortably on your desk.
- The printer should come bundled with the requisite cables, or you may have to spend more purchasing them separately.